

KHETHAVATH SUNIL NAIK

Data Science & Machine Learning | Full-Stack Development

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Final-year B.Tech student in Data Science & AI at IIT Bhilai with hands-on experience deploying adaptive machine learning systems in live school environments. Built and shipped an ELO- and graph-based adaptive learning platform and a momentum-aware assessment algorithm (paper submitted to IEEE DSAA 2026), backed by full-stack engineering skills to take ML systems from prototype to production.

EDUCATION

Indian Institute of Technology (IIT) Bhilai 2023 – 2027
B.Tech in Data Science & Artificial Intelligence — CGPA: 7.66/10

EXPERIENCE

Apprentice — ShikshaSarathi May 2026 – Jul 2026
Under Dr. Gagan Raj Gupta, Associate Professor, CSE, IIT Bhilai

- Built the complete offline-first school learning & quiz platform end-to-end and converted it into a deployable app, rolled out across multiple schools.
- Implemented the MARS (Momentum-Aware Adaptive Rating System) algorithm as the platform's adaptive-testing engine, powering real-time learner rating and question routing.

Data Analyst Intern — Unified Mentor Apr 2025 – Aug 2025

- Cleaned and analyzed 10,000+ records using Python (Pandas) and SQL, uncovering customer segments that informed business strategy.
- Built Tableau dashboards automating reporting, saving 3+ hours/week; identified churn patterns via cohort analysis for stakeholder presentations.

Robotics & AI Intern — RoboAI Nov 2024 – Jan 2025

- Designed CAD models and implemented pathfinding algorithms (A*, Dijkstra), improving automation efficiency by 15%.
- Integrated computer-vision modules into robotic pipelines, accelerating object detection in constrained hardware environments.

PUBLICATION

MARS: Momentum-Aware Adaptive Rating System Submitted, IEEE DSAA 2026

- Designed a graph-based adaptive assessment algorithm combining momentum-aware rating updates, response-quality scoring, and time-budgeted routing.
- Deployed in production within ShikshaSarathi and GATEWay.

KEY PROJECTS

GATEWay — Adaptive Learning Platform 2026

- Architected ELO-based skill tracking, weak-topic detection, and a graph-based recommendation engine, reducing study redundancy by 40% in pilot tests.
- Integrated Supabase backend with real-time analytics, OAuth2 auth, and a React + Node.js frontend achieving sub-200ms query response times.

AgriBot — Weed Detection System 2024

- Built a CNN-based weed detection pipeline (YOLOv11n + Intel RealSense D435i) achieving ~92% mAP@50, deployed on a physical rover via a Flask API for real-world field use.

EmoCare — Mental Health Detection via NLP 2025

- Built a multi-model NLP pipeline (BERT, CNN, LSTM) achieving 88%+ accuracy on mental health benchmarks, with a Streamlit chatbot interface.

Additional: Pacman (A-driven ghost AI, Python) · TCS stock forecasting (ARIMA + LSTM, 18% lower RMSE) · Uber peak-demand analytics (Plotly, Power BI)*

TECHNICAL SKILLS

AI / ML: PyTorch, TensorFlow, Scikit-learn, BERT, CNNs, LSTMs, Transformers, Computer Vision, NLP

Languages: Python, C, C++, SQL, JavaScript, Bash

Data & Visualization: Pandas, NumPy, Plotly, Tableau, Power BI, Excel

Databases: MySQL, PostgreSQL, MongoDB, Supabase, Hadoop, Spark

Full Stack & Tools: React, Node.js, Express.js, REST APIs, Flask, Streamlit, Git, Docker

LEADERSHIP & ADDITIONAL

- Operations Head, DesignX, IIT Bhilai (Jul 2025 – Apr 2026); Head of 3D Modeling (Jul 2024 – Apr 2025)
- Founder, Kalashetra Fine Art Auctions — live full-stack art auction platform (thekalashetra.vercel.app), supported by India Art Fair
- Media Head, FinTech, IIT Bhilai (Aug 2024 – May 2025)